

exness_bot — Step-by-Step Guide

LLM + rule-based auto-trading bot for Exness (MetaTrader 5)

Version 1.2 · Author: Waleed Hassan · Runs on Windows 7, 10 and 11

Read this first. Automated FX/CFD trading loses money for most retail traders. This bot is a **framework and starting point** and has **not** been run against a live account by the author. Always go **backtest** → **demo for weeks** → **tiny live size**, in that order. Algo trading is allowed by Exness; arbitrage, latency exploitation and quote-delay tick-scalping are not. Nothing here is financial advice.

Requirements (full list)

To backtest only (any OS, no broker needed)

Need	Detail
Operating system	Windows, macOS or Linux
Python	3.8–3.12, 64-bit (tick <i>Add Python to PATH</i>)
Python packages	<code>pandas</code> , <code>numpy</code> — <code>pip install pandas</code> pulls both
Data	one CSV of candles, columns <code>time, open, high, low, close</code>

To trade (demo or live)

Need	Detail
Operating system	Windows 7 SP1 / 10 / 11, 64-bit. The <code>MetaTrader5</code> package is Windows-only; a Windows VPS also counts.
Python	Windows 7 → Python 3.8.10 (last version that installs on Win 7). Windows 10 / 11 → latest Python 3. 64-bit, <i>Add Python to PATH</i> .
Python packages	<code>MetaTrader5>=5.0.37</code> , <code>pandas>=1.3.5</code> , <code>numpy>=1.21</code> ; plus <code>openai>=1.10,<2</code> only for LLM mode. All installed by <code>pip install -r requirements.txt</code> .
Broker terminal	Exness MetaTrader 5 desktop terminal, installed and logged in.
Exness account	Start with a demo account — you need its <i>login number</i> , <i>password</i> and <i>server</i> (e.g. <code>Exness-MT5Trial</code>).
Terminal setting	Tools → Options → Expert Advisors → Allow algorithmic trading ticked.
Network / uptime	Stable internet; the PC or VPS stays on and the terminal stays open while the bot runs.
Disk / RAM	~1 GB free disk (Python + MT5), 2 GB+ RAM.
Optional	An OpenAI API key — only if <code>USE_LLM = True</code> and you want LLM decisions instead of the built-in rules.

Hardware needs are light: the bot polls every few seconds and holds one position at a time. Any machine that runs the MT5 terminal comfortably runs the bot.

Part A — The easy way (no coding needed)

If you are not a programmer, follow just this part. You need a **Windows PC** (7, 10 or 11), a **free Exness demo account**, and about **20 minutes**. You will not type any code — only double-click a few files and fill in one form.

1. Step 1 — Install Python (one time)

Go to `python.org/downloads`.

- **Windows 7:** download **Python 3.8.10** (page: `python.org/downloads/release/python-3810/` → "Windows installer 64-bit"). Newer Python versions do *not* install on Windows 7.
- **Windows 10 / 11:** download the latest Python 3.

Run the installer.

Tick the box "Add Python to PATH"

, then click *Install Now*.

2. Step 2 — Download the bot

Open `github.com/waleed260/exness-bot`. Click the green

Code

button →

Download ZIP

. Right-click the downloaded ZIP → *Extract All*. Remember where the folder is (e.g. your Desktop).

3. Step 3 — Install MetaTrader 5 from Exness

In your Exness Personal Area, create a **demo**

account and note its *login number*, *password* and *server* (e.g. `Exness-MT5Trial`). Download and install the

Exness MetaTrader 5

terminal and log into that demo account. In MT5:

Tools → **Options** → **Expert Advisors**

→ tick

"Allow algorithmic trading"

→ OK.

4. Step 4 — Set it up (one time)

Open the extracted bot folder. Double-click

`install.bat`

. A black window installs everything (a few minutes), then a small text file opens in Notepad. Fill in your demo *login number*, *password* and *server* between the quote marks, then

Save

(**Ctrl** + **S**) and close Notepad.

If it says "Python is not installed", you missed the "Add Python to PATH" tick in Step 1 — re-run the Python installer and choose *Modify*.

5. Step 5 — Run it

Double-click

run.bat

. A black window opens and starts printing what the bot is doing. It is in **safe DRY-RUN mode**

: it makes decisions and writes them down but **does not place any real orders**

. Leave it running as long as you like.

To stop: click the window and press **Ctrl** + **C**, or just close the window.

6. Step 6 — See what it did

Inside the bot folder open `exness_bot\logs\`: `bot.log` is the full running commentary; `trades.csv` lists every trade it opened/closed (open it in Excel).

7. Step 7 — Test on past data (optional but recommended)

In MetaTrader 5: *View* → *Symbols* → *Bars*, pick your pair and timeframe, *Export Bars* to a CSV file. Then double-click

backtest.bat

and drag that CSV into the window. It prints whether the strategy would have made or lost money, and a plain

EDGE / NO EDGE

verdict.

8. Step 8 — Only when you are ready for more

- *Trade the demo for real* (still fake money): open `exness_bot\config.py` in Notepad, find `DRY_RUN = True`, change it to `DRY_RUN = False`, save. Run `run.bat` again.
- *Real money*: only after several weeks of good demo results. See Part 6. Start with the smallest possible size.

Golden rules. 1) Never skip the demo stage. 2) Never use money you cannot afford to lose. 3) The built-in strategy is a *starting point*, not a money machine — prove it with the backtester and the demo first.

The rest of this document (Parts 1–9) is the detailed reference: what each part does, every setting, command-line usage, an honest assessment for Exness, and troubleshooting.

1. What the bot does

Once per **closed candle** (default timeframe M15) it runs this pipeline:

```
MT5 price history
  ▼
indicators  SMA(20), SMA(50), RSI(14), ATR(14), 200-SMA trend
  ▼
strategy    LLM decision (OpenAI) --falls back to--> rule set
                                                    (SMA crossover + RSI + trend)
  ▼
decision    buy / sell / close / hold (+ confidence 0..1)
  ▼
filters     session window? spread ok? min ATR? confidence >= min?
            daily loss limit not hit? no position already open?
  ▼
risk        lot size from RISK_PER_TRADE_PCT
            SL = SL_ATR_MULT x ATR, TP = TP_ATR_MULT x ATR
  ▼
executor    mt5.order_send (retries on requote) [skipped if DRY_RUN]
```

On **every poll** (default every 5 s) it also manages any open position:

- at **+1.0R** profit → move the stop loss to entry (break-even);
- at **+1.5R** profit → trail the stop **2 × ATR** behind price;
- the stop is only ever tightened, never loosened.

R = the initial risk distance (entry → stop loss). All activity is logged to `exness_bot/logs/bot.log`; each entry/exit is appended to `exness_bot/logs/trades.csv`.

2. What every file is for

File	Job
<code>exness_bot/config.py</code>	all settings. <code>DRY_RUN=True</code> , <code>DEMO_ONLY=True</code> by default
<code>exness_bot/settings.example.py</code>	copy to <code>settings.py</code> ; MT5 login + optional OpenAI key (git-ignored)
<code>exness_bot/mt5_client.py</code>	connect, account, positions, candles; auto-resolves the broker symbol suffix; reports spread
<code>exness_bot/indicators.py</code>	indicator maths + trailing-stop + session logic. No MT5 import, so the backtester reuses it
<code>exness_bot/data.py</code>	pulls live candles → DataFrame with indicator columns; drops the still-forming candle
<code>exness_bot/strategy.py</code>	builds the prompt, calls the LLM, validates the JSON; rule-based fallback
<code>exness_bot/risk.py</code>	lot sizing, SL/TP, break-even + trailing stop, daily-loss guard, session check
<code>exness_bot/executor.py</code>	<code>mt5.order_send</code> for open / close / modify-stop, with retries; obeys <code>DRY_RUN</code>
<code>exness_bot/runner.py</code>	the main loop that ties it together
<code>exness_bot/backtest.py</code>	runs the rule strategy over history and prints edge metrics
<code>install.bat</code> / <code>run.bat</code> / <code>backtest.bat</code>	Windows double-click helpers (Part A) — they just call the commands below for you

3. Every setting in `config.py`

Safety

Setting	Default	Meaning
DRY_RUN	True	True = never send a real order, only log what it would do
DEMO_ONLY	True	refuse to start if the connected account is a live account
CONFIRM_LIVE_STRING	""	must equal "I ACCEPT THE RISK" to run on a live account

Market

Setting	Default	Meaning
SYMBOL	EURUSD	base name; broker suffix (m , . , z ...) found automatically
TIMEFRAME	M15	candle used for decisions
LOOKBACK_BARS	400	candles pulled for indicators / LLM context

Risk

Setting	Default	Meaning
RISK_PER_TRADE_PCT	0.5	percent of balance lost if the stop is hit
FIXED_LOT_FALLBACK	0.01	lot used if sizing can't be computed
MAX_LOT	0.10	hard cap on lot size
MAX_OPEN_POSITIONS	1	positions allowed at once on this symbol
ATR_PERIOD	14	ATR length
SL_ATR_MULT	2.0	stop distance = this × ATR
TP_ATR_MULT	3.0	target distance = this × ATR
MIN_ATR_POINTS	0	skip entries when ATR (points) is below this; 0 = off
DAILY_MAX_LOSS_PCT	3.0	stop opening trades for the day after this equity drawdown
MIN_CONFIDENCE	0.55	ignore buy/sell signals weaker than this

Trailing / break-even

Setting	Default	Meaning
BREAKEVEN_AT_R	1.0	move stop to entry once profit reaches this many R; 0 = off
TRAIL_AT_R	1.5	start trailing once profit reaches this many R; 0 = off
TRAIL_ATR_MULT	2.0	trailing stop sits this × ATR behind price

Entry filters

Setting	Default	Meaning
MAX_SPREAD_POINTS	25	skip entries when the spread is wider than this
USE_TREND_FILTER	True	only long in an up-trend, only short in a down-trend
TREND_SMA	200	SMA used as the trend proxy
TREND_SLOPE_LOOKBACK	20	bars back used to measure the trend SMA slope
SESSION_UTC_HOURS	[7, 16]	trade only between these UTC hours; [] = always
TRADE_DAYS	[0, 1, 2, 3, 4]	weekdays allowed (Mon–Fri)

Execution

Setting	Default	Meaning
DEVIATION_POINTS	20	maximum slippage accepted, in points
MAGIC	20240601	id stamped on this bot's orders
POLL_SECONDS	5	how often the loop checks for a new candle / manages the stop
ORDER_RETRIES	3	retries on requote / price-changed
TRADE_LOG_CSV	exness_bot/logs/trades.csv	where trades are recorded

Strategy

Setting	Default	Meaning
USE_LLM	True	use the LLM if an API key is set, else the rule set
SMA_FAST / SMA_SLOW	20 / 50	crossover SMAs
RSI_PERIOD	14	RSI length

Backtest

Setting	Default	Meaning
BT_SPREAD_POINTS	12	spread cost (points) charged on every backtested trade
BT_COMMISSION_PER_LOT	7.0	informational commission assumption for Raw/Zero accounts
BT_START_BALANCE	1000	starting balance for the money curve

4. Step-by-step: backtest (do this first)

Runs on **any OS** — no MetaTrader, no Windows needed.

1. Install Python 3.10+ and pandas:

```
pip install pandas
```

2. Get historical candles as a CSV with columns `time`, `open`, `high`, `low`, `close` (export from MT5: *View* → *Symbols* → *Bars*, or any data provider). Save as e.g. `data/EURUSD_M15.csv`.
3. Run:

```
python -m exness_bot.backtest --csv data/EURUSD_M15.csv
```

On Windows with MT5 connected you can pull data directly instead:

```
python -m exness_bot.backtest --mt5 180
```

4. Read the report:

```
trades      : 15
win rate    : 26.7%
expectancy  : -0.604R per trade
profit factor : 0.30
total       : -9.1R
max drawdown : 4.5%
verdict     : NO EDGE - do not trade this as-is
```

- **expectancy > 0** and **profit factor > 1.15** → maybe an edge, forward-test it.
- otherwise → change the strategy or parameters and run again.

5. Per-trade detail is written to `exness_bot/logs/backtest_trades.csv`.

Never move to live money on a “NO EDGE” result.

5. Step-by-step: demo on Windows

The `MetaTrader5` Python package only runs on **Windows** (a Windows VPS is fine).

Python version: on **Windows 7** use **Python 3.8.10** — the last release that installs on Win 7, and `requirements.txt` is set so the right package versions install automatically. On **Windows 10 / 11** use the latest Python 3. Either way tick "Add Python to PATH" and use the **64-bit** installer.

1. Create a **demo** account in the Exness Personal Area; note the login number, password and server (e.g. `Exness-MT5Trial`).
2. Install the **Exness MetaTrader 5** terminal, log into the demo account.
3. In the terminal: **Tools** → **Options** → **Expert Advisors** → **Allow algorithmic trading** (tick it).
4. Install dependencies:

```
pip install -r requirements.txt
```

5. Create your private config:

```
copy exness_bot\settings.example.py exness_bot\settings.py
```

Edit `exness_bot\settings.py`:

```
MT5_PATH      = r"C:\Program Files\MetaTrader 5 EXNESS\terminal64.exe"
MT5_LOGIN     = 12345678
MT5_PASSWORD  = "your-demo-password"
MT5_SERVER    = "Exness-MT5Trial"
OPENAI_API_KEY = ""           # leave empty to use the rule-based strategy
OPENAI_MODEL  = "gpt-4o-mini"
```

(`settings.py` is git-ignored — credentials never get committed.)

6. First run with `DRY_RUN = True` (the default):

```
python -m exness_bot.runner
```

Watch `exness_bot/logs/bot.log`. You'll see decisions and lines like `[DRY_RUN] would OPEN buy 0.02 EURUSD ...`. No orders are sent.

7. When the dry-run behaviour looks sane, set `DRY_RUN = False` in `exness_bot/config.py` and run again. Now it trades the **demo** account for real. Let it run for **several weeks**; review `logs/trades.csv`.

6. Step-by-step: going live (only after profitable demo)

1. In `exness_bot/config.py`:

```
DEMO_ONLY = False
CONFIRM_LIVE_STRING = "I ACCEPT THE RISK"
MAX_LOT = 0.01
RISK_PER_TRADE_PCT = 0.25      # start smaller than on demo
```

2. Put your **live** account details in `settings.py`.
3. Run `python -m exness_bot.runner`. It logs a warning that it is on a live account.
4. Watch it daily for the first weeks. Keep the daily loss limit tight.
5. Scale size up only after months of consistent results, never faster than your drawdown tolerance.

Keep the machine/VPS on and the MT5 terminal running. If the terminal closes, the bot loses its connection — open positions keep their broker-side SL/TP, but the trailing logic pauses until it reconnects.

7. Run it in VS Code (for developers)

1. Install VS Code

and the

Python extension

(Microsoft).

2. File → Open Folder...

and pick the extracted `exness-bot` folder.

3. Terminal → New Terminal

to get a shell at the project root.

4. (Recommended) create a virtual environment so packages stay local:

```
python -m venv .venv
.venv\Scripts\activate      # Windows
# source .venv/bin/activate  # macOS / Linux (backtest only)
```

Then click the Python version in the status bar (bottom-right) and select the `.venv` interpreter.

5. Install packages:

```
pip install -r requirements.txt    # trading (Windows)
pip install pandas                # backtest only (any OS)
```

6. Create the settings file once:

```
copy exness_bot\settings.example.py exness_bot\settings.py    # Windows
```

Open `exness_bot/settings.py` and fill in the demo login / password / server.

7. Run from the VS Code terminal:

- Backtest — `python -m exness_bot.backtest --csv data/EURUSD_M15.csv`
- Bot (safe DRY-RUN) — `python -m exness_bot.runner`
- Or open `exness_bot/runner.py` and press **F5** to run under the debugger (set breakpoints in `strategy.py` / `risk.py` to watch a decision form).

8. Stop the bot with **Ctrl+C** in the terminal.

Output goes to `exness_bot/logs/bot.log` and `exness_bot/logs/trades.csv` — open them in the editor while it runs.

8. Deploy — keep it running 24/5

The bot only works while it is running **and** the Exness MT5 terminal is open and logged in. Pick one:

Option A — your own PC (simplest). Keep the PC on with MT5 running and `run.bat` (or `python -m exness_bot.runner`) going in a window. *Settings* → *Power & sleep* → *Sleep* = *Never* so it does not suspend.

Option B — a Windows VPS (recommended for real use).

1. Rent a small **Windows Server VPS** (~2 vCPU / 4 GB RAM is plenty; "Forex VPS" plans work well and sit close to the broker).
2. Connect with **Remote Desktop** (`mstsc`).
3. On the VPS: install Python (same version rule as Part A / section 5), install the **Exness MT5** terminal, log in, tick *Allow algorithmic trading*.
4. Copy the `exness-bot` folder across, run `install.bat`, fill in `settings.py`.
5. Start it with `run.bat`. It keeps running after you close the Remote Desktop window — just don't log out of the Windows session.
6. Auto-start after a reboot: put a shortcut to `run.bat` in `shell:startup`, and set the MT5 terminal to start with Windows.

Keep it alive unattended. Run the bot under a supervisor that restarts it if it exits — e.g. **NSSM** (`nssm install exness-bot` → command `python -m exness_bot.runner`) to register it as a Windows service, or Task Scheduler with *Restart on failure*. Check `logs/bot.log` daily for the first week.

No cloud / Docker option. `MetaTrader5` needs a real Windows session with the terminal running, so a Linux container cannot trade. A Linux box is fine for backtesting only.

9. How good is it for Exness — honest assessment

a) Will it work *technically* on Exness?

Fairly likely after a short debugging pass on demo. Usual first-run snags:

Snag	Fix
wrong <code>MT5_SERVER</code> string	<code>initialize failed</code> → copy the exact server name from the terminal
symbol is <code>EURUSDm</code> / <code>EURUSD.z</code> on your account	handled automatically; check the log line <code>Resolved symbol EURUSD -> ...</code>
"AutoTrading disabled"	step 5.3 above
broker min stop distance > your ATR stop	the bot clamps it; on very tight timeframes SL/TP get pushed out — use a higher timeframe

Rough chance of a clean demo run within a day or two: **~85%**.

b) Will it make real profit?

The built-in strategy (SMA crossover + RSI + trend filter, or a generic LLM prompt) is a **baseline, not an edge**. After spread, swap and commission its expected value is around zero-to-negative. Realistic odds that this exact configuration is net-profitable over a year: **~10–15%**, in line with retail algo-trading outcomes generally.

What actually moves that number:

- A strategy with a tested statistical edge — validate with `backtest.py` over several years and multiple symbols, then walk-forward.
- Trading the right sessions / instruments for that edge.
- Costs: a low-spread account type; avoid holding across the 3-day swap.
- Risk discipline: small `RISK_PER_TRADE_PCT`, a hard daily stop — which this bot already enforces.

Bottom line: treat this as a solid *execution and risk* framework with a *backtester* attached. The profitable idea has to be yours and has to survive the backtest before any money is at stake.

10. Troubleshooting

Symptom	Cause / fix
<code>settings.py</code> not found	you didn't copy <code>settings.example.py</code> → <code>settings.py</code>
<code>mt5.initialize</code> failed	wrong <code>MT5_PATH</code> , <code>MT5_SERVER</code> , login or password; terminal not installed
Symbol <code>EURUSD(+suffix)</code> not found	your account uses a different name; add its suffix to <code>_SUFFIXES</code> in <code>mt5_client.py</code>
Decisions logged but no orders	<code>DRY_RUN</code> is still <code>True</code>
<code>OPEN rejected: retcode=10027</code>	AutoTrading disabled in the terminal
<code>OPEN rejected: retcode=10016</code>	invalid stops (broker min distance); raise <code>SL_ATR_MULT</code> or use a higher timeframe
<code>OPEN rejected: retcode=10019</code>	not enough money for that lot; lower <code>RISK_PER_TRADE_PCT</code> or <code>MAX_LOT</code>
No trades ever	trend filter + session window too strict, or <code>MIN_CONFIDENCE</code> too high; loosen and re-backtest
LLM errors in the log	bad/empty <code>OPENAI_API_KEY</code> ; the bot auto-falls back to the rule set
<code>ModuleNotFoundError: MetaTrader5</code> on Win 10/11	you installed 32-bit Python, or Python is too new for a wheel — use the 64-bit installer and re-run <code>install.bat</code>
Python installer says "not supported on this OS" (Win 7)	you downloaded a version newer than 3.8.10 — get Python 3.8.10

11. Safe-launch checklist

- ☐ Backtested on ≥ 2 years of data, on ≥ 2 symbols → positive expectancy, $PF > 1.15$
- ☐ Walk-forward / out-of-sample check also positive
- ☐ Ran on demo with `DRY_RUN=True` , decisions look sane in the log
- ☐ Ran on demo with `DRY_RUN=False` for ≥ 4 weeks → result matches the backtest
- ☐ `RISK_PER_TRADE_PCT` ≤ 0.5 , `DAILY_MAX_LOSS_PCT` ≤ 3 , `MAX_LOT` small
- ☐ Account type has low spread; you understand swap on your instrument
- ☐ VPS / machine stays on; MT5 terminal set to start with the OS
- ☐ You can afford to lose the whole account balance

Only when every box is ticked: set the two live flags and start at `0.01` lots.